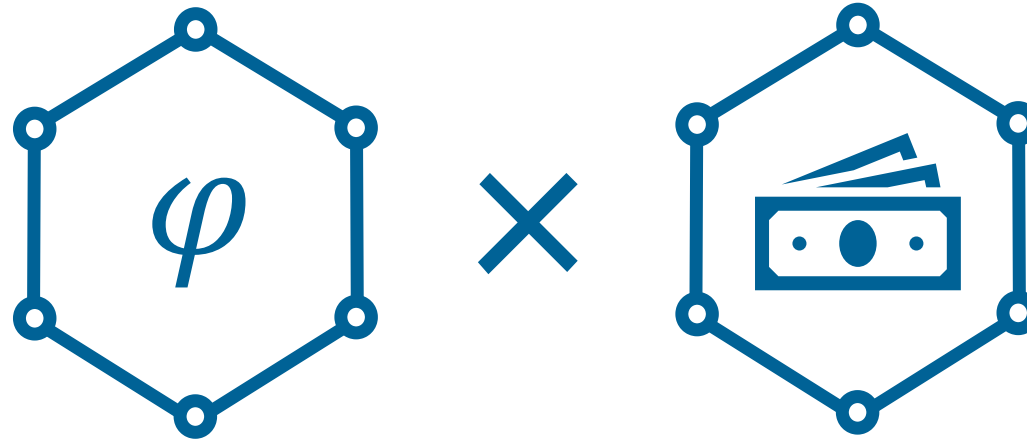
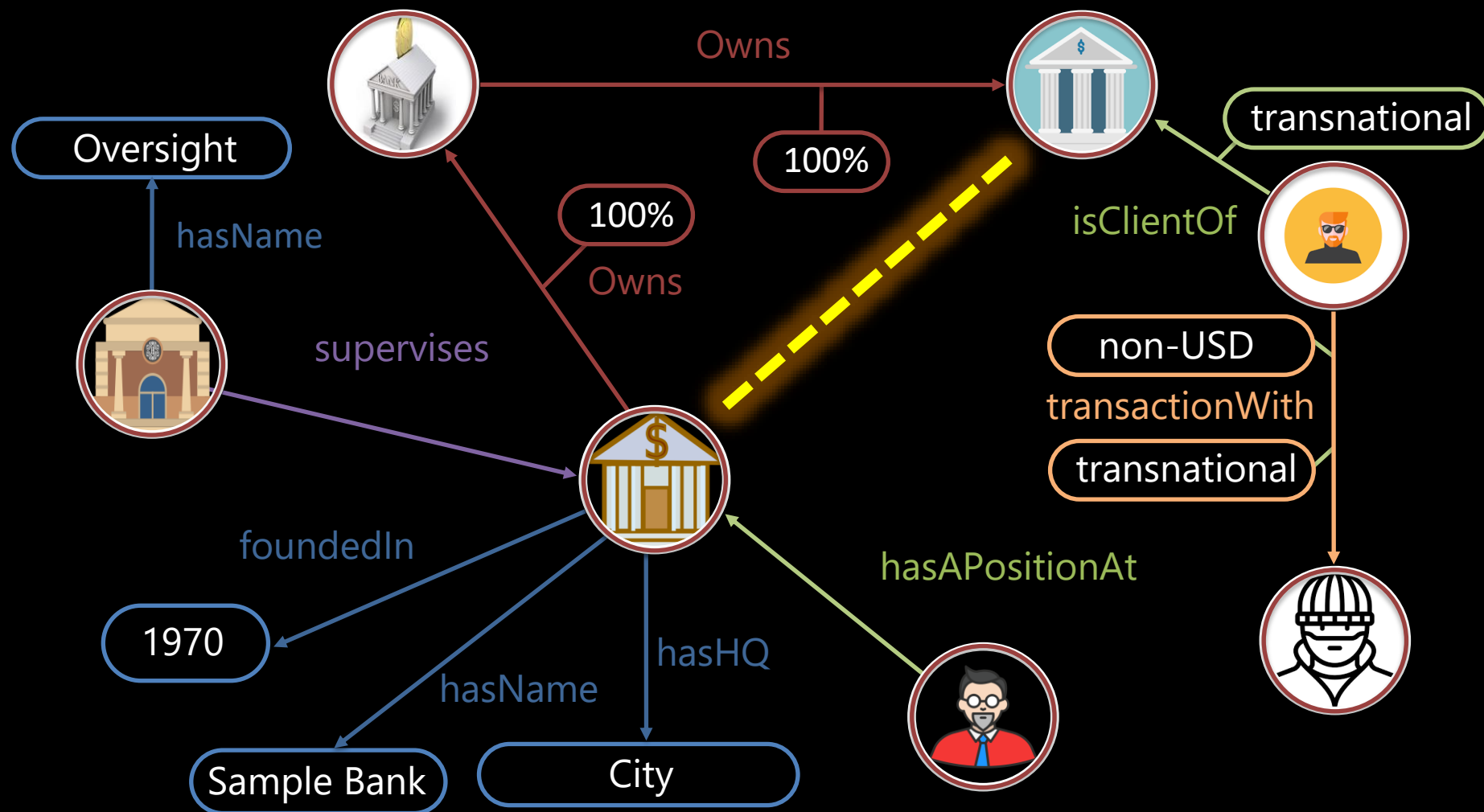
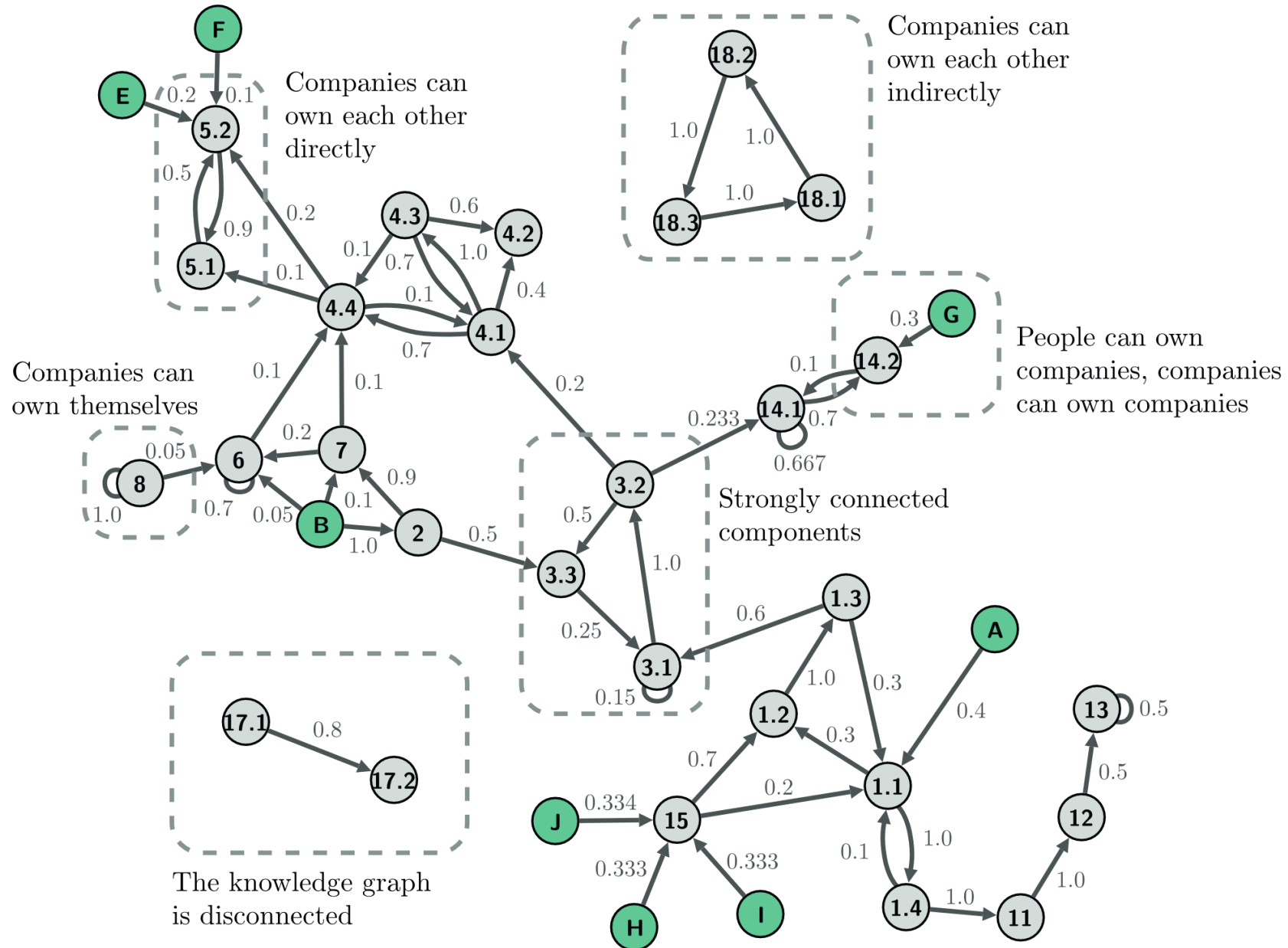


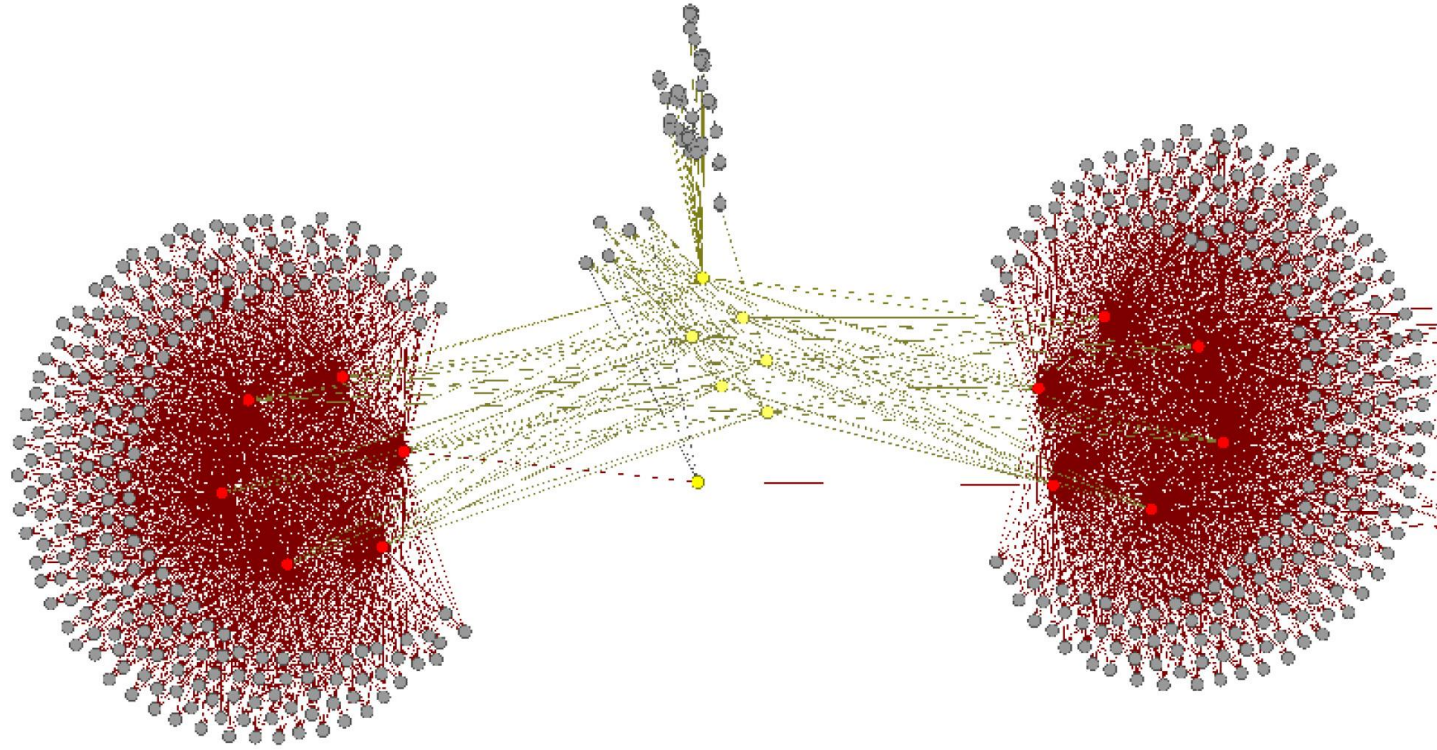


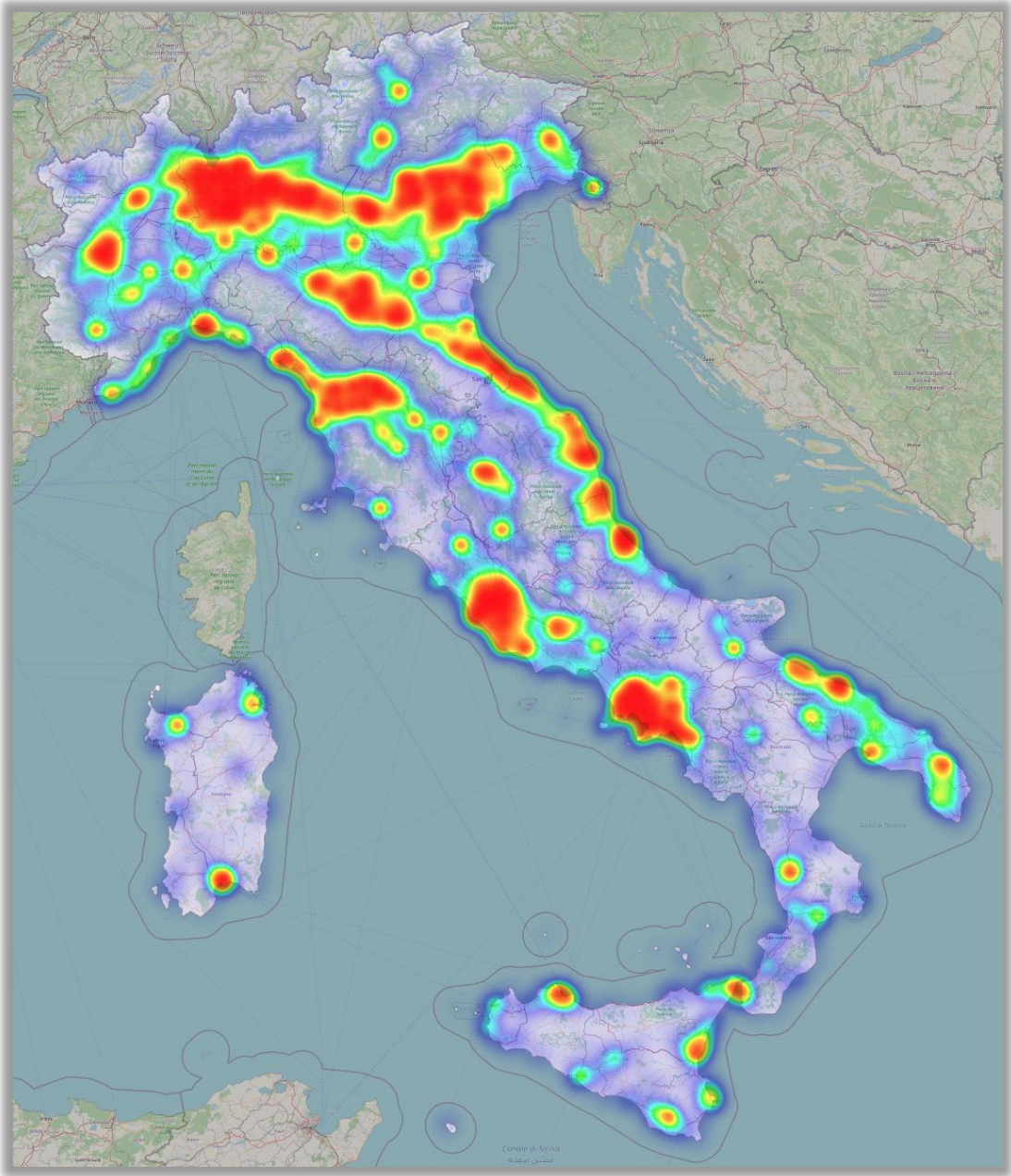
Logical Knowledge in KGs The Real World

Emanuel Sallinger









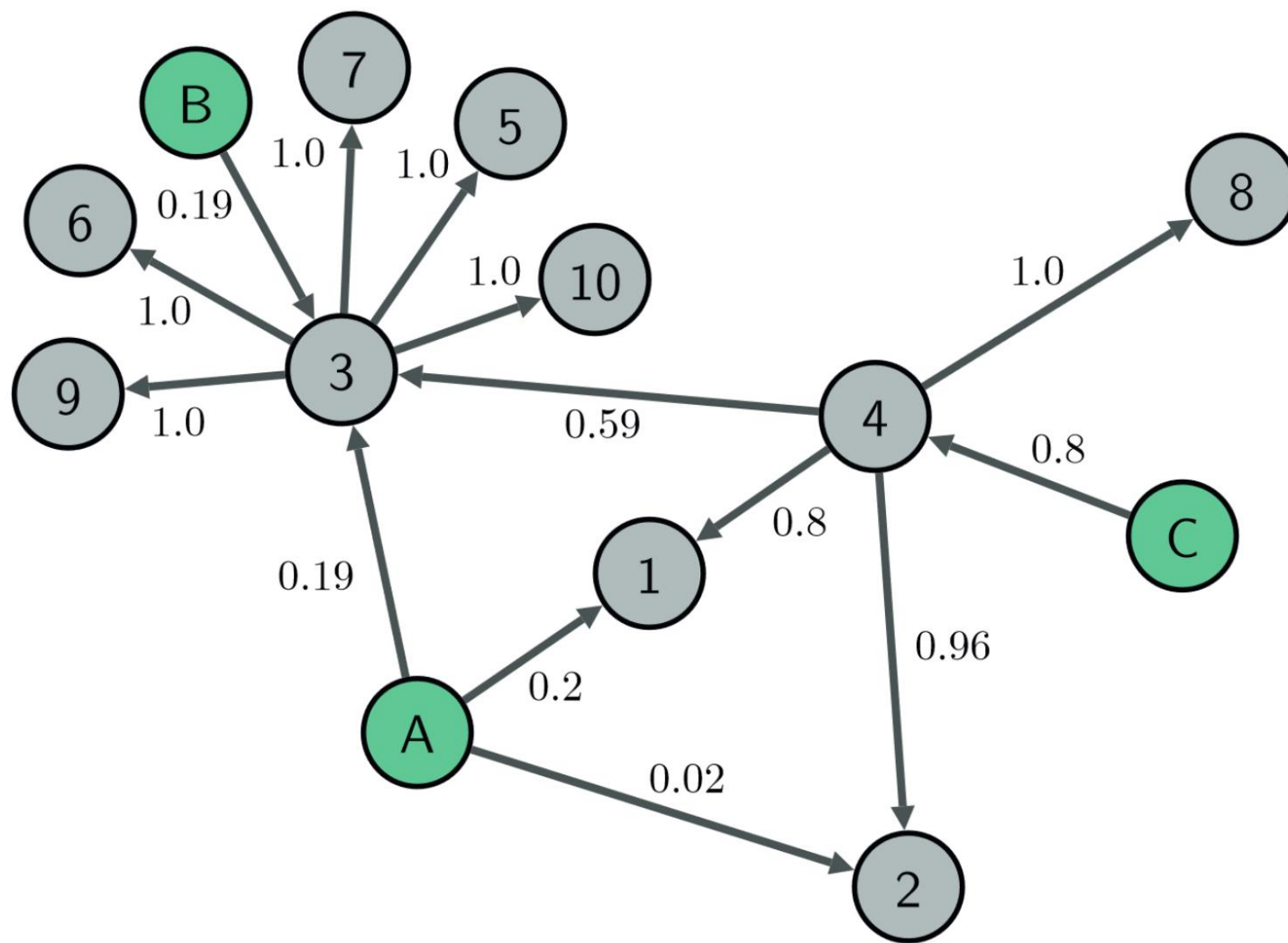


Conglomerates





Conglomerates



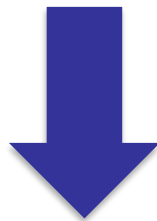


Conglomerates

Definition 5. *The undirected ϵ -Baldone ownership of a company s on a company t in a graph G is a function $\mathcal{U}_\epsilon^G(s, t) : \max\{O_\epsilon^G(s, t), O_\epsilon^G(t, s)\}$.*

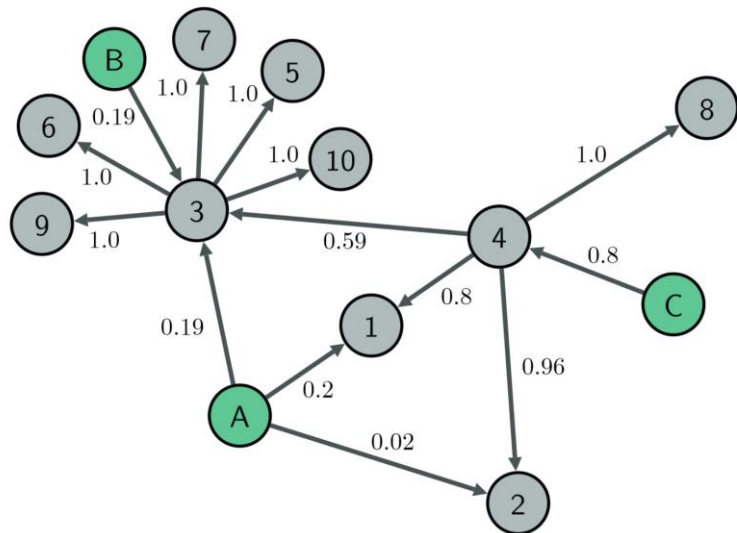
Definition 6. *The binary relation vicinity $\mathcal{V}_\epsilon^G$ holds between two companies x and y of a graph G (and we say — they are close —), if: (i) $\mathcal{U}_\epsilon^G(x, y) > 0$; or, (ii) there exists a third party z of G such that $\mathcal{U}_\epsilon^G(z, x) > 0$ and $\mathcal{U}_\epsilon^G(z, y) > 0$.*

Definition 7. *An ϵ -conglomerate is an equivalence class of $\mathcal{V}_\epsilon^{+G}$, where $+$ denotes the transitive closure of the vicinity relation defined on a graph G .*





Conglomerates



$$O(a, b, x), x > \epsilon \rightarrow U(a, b, x), U(b, a, x) \quad (1)$$

$$U(a, b, x), a > b \rightarrow \exists z \ C(z, a), C(z, b) \quad (2)$$

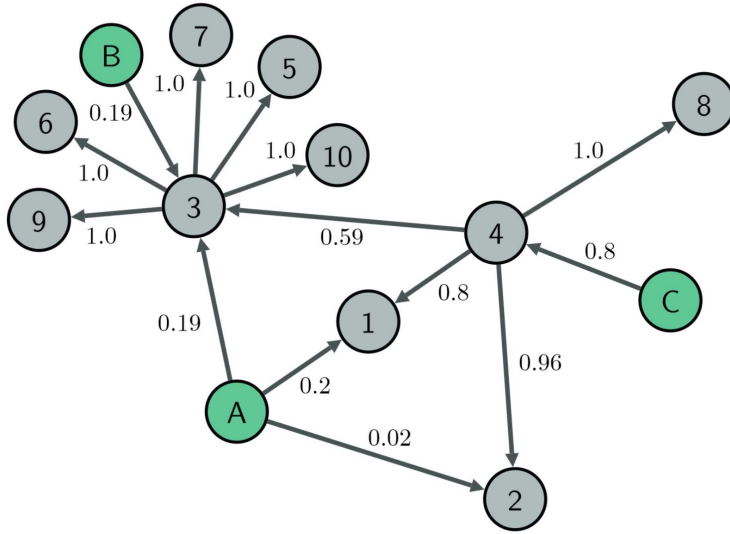
$$U(w, a, x), U(w, b, x) \rightarrow \exists z \ C(z, a), C(z, b) \quad (3)$$

$$Company(a) \rightarrow \exists z \ C(z, a) \quad (4)$$

$$C(y, a), C(x, a) \rightarrow x = y \quad (5)$$



Conglomerates



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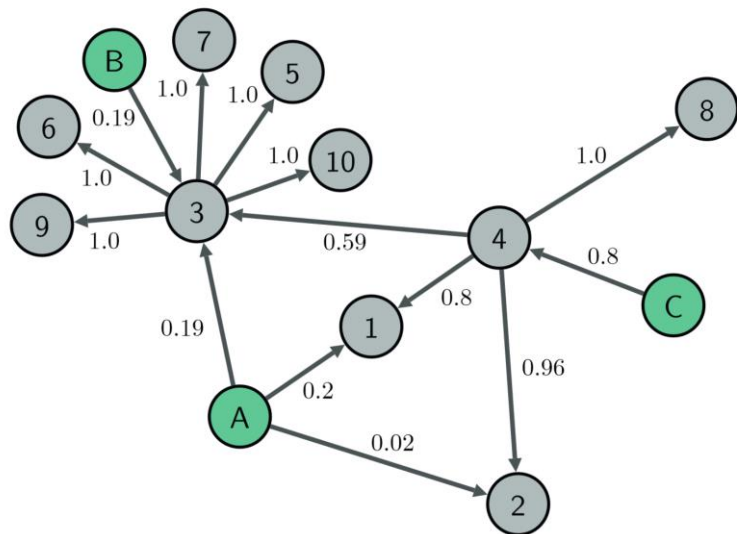
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Collusion





Considering Collusion

Collusion Golden Power Check

- Goal:* The goal is checking whether an acquisition (of shares, stocks, etc.) causes any strategic Italian company to be possibly controlled by a set of foreign companies acting in collusion.
- Setting:* Let S be a set of strategic companies and F be a set of foreign companies. Let t be a transaction (e.g., an offer issued by a company x to buy an amount s of shares of a company y), with $x, y \in S \cup F$.
- Question:* Decide whether t causes F to jointly control a company in S .
- Insight:* Consider the possibility to exert Golden Powers to block t .



Considering Collusion

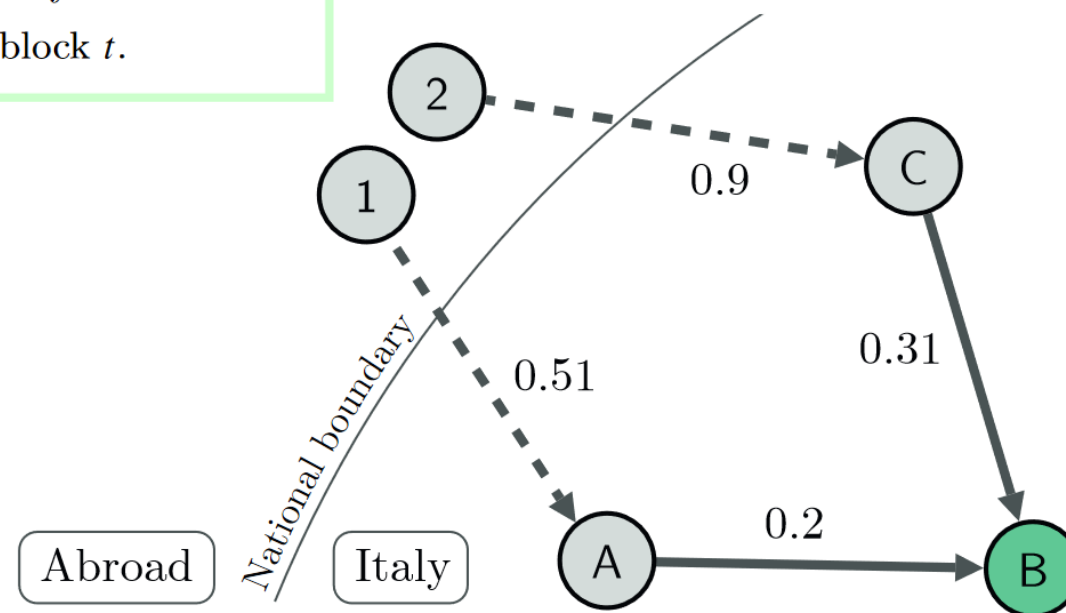
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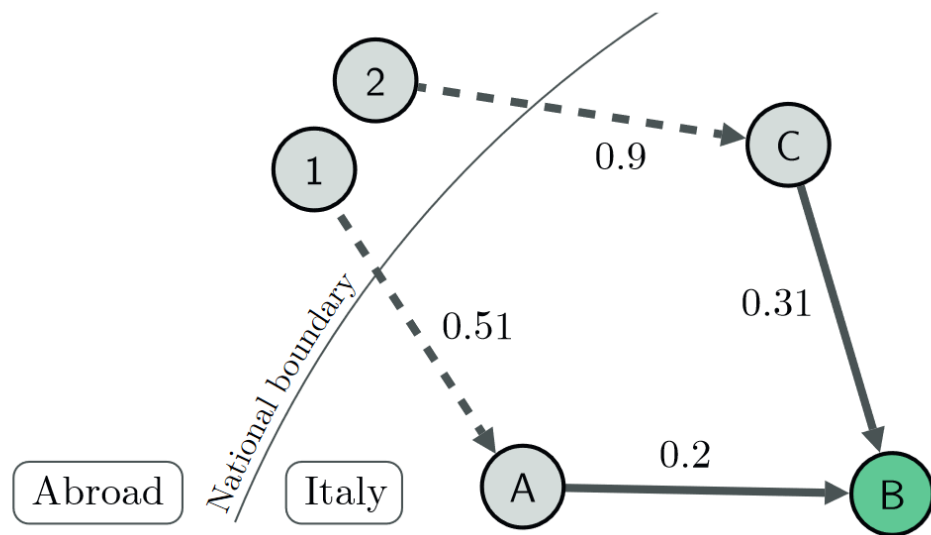
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Considering Collusion



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$$T(x, y, w) \rightarrow \textit{Own}(x, y, w) \quad (2)$$

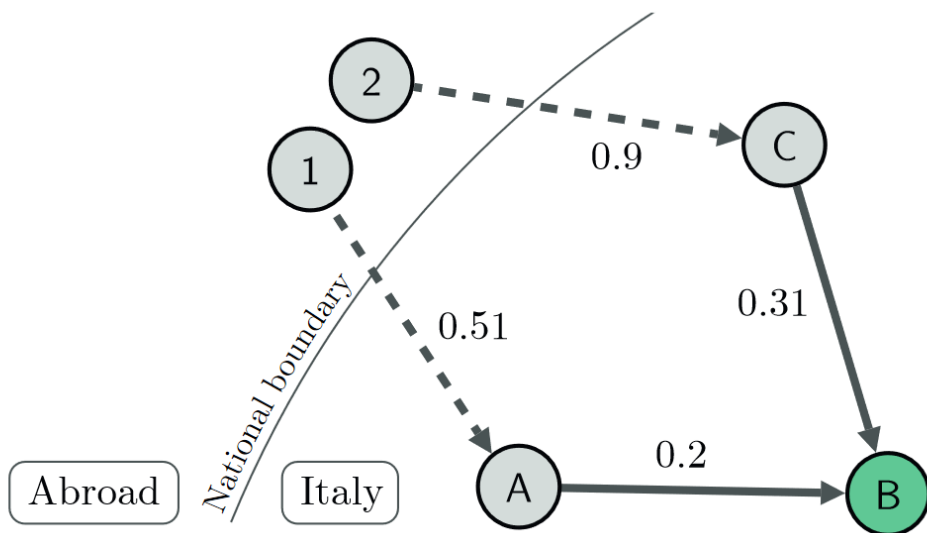
$$\textit{Control}(x, y) \rightarrow \textit{Control}(x, x) \quad (3)$$

$$\textit{Control}(x, y), \textit{Own}(y, z, w), v = \textit{msum}(w, \langle y \rangle), v > 0.5 \rightarrow \textit{Control}(x, z) \quad (4)$$

$$F(x), S(y), \textit{Control}(x, y) \rightarrow \textit{CGPCheck}(x, y) \quad (5)$$



Considering Collusion



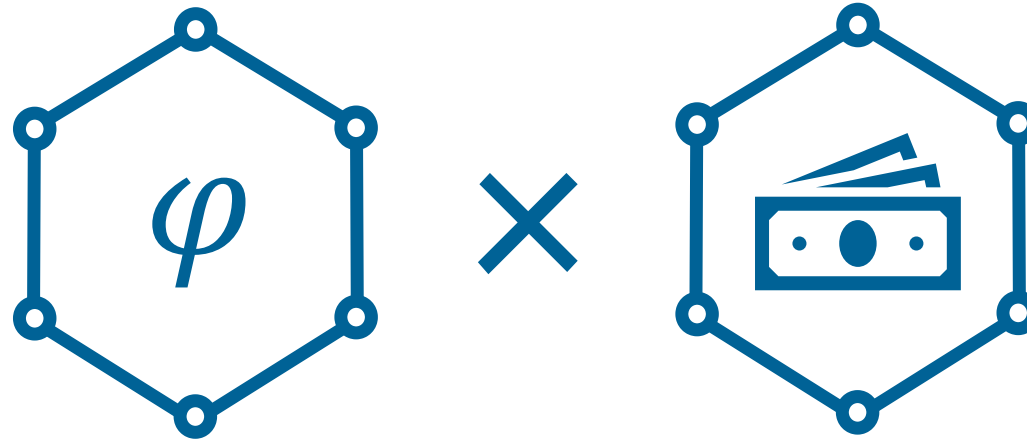
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